The University of Sydney

BIOL2022
Biology Experimental Design and Analysis

BEDA

Unit Handbook

Dr. Januar Harianto
Prof. Clare McArthur
Prof. Matthew Crowther

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About this edition

This edition of the *Biology Experimental Design and Analysis Unit Handbook* was prepared for BIOL2022 at the University of Sydney.

Biology Experimental Design and Analysis changes as we refine the teaching, practical activities, datasets and assessments. Students returning to this material should use the edition from the year in which they completed the unit. Later editions may differ, sometimes substantially.

Authors

Januar Harianto · Clare McArthur · Matthew Crowther

How this book was built

This handbook is written in Quarto using Markdown and R. The online handbook is published as HTML, while the A4 edition is typeset with Typst. The print edition uses Libertinus Serif throughout. The source is developed openly on GitHub, and meaningful changes are recorded on the Updates page.

Rendering the handbook

Most students should use the published PDF. Students who want to build a local copy can install Quarto 1.9 or later and R. The rendering instructions in the source repository explain the remaining steps.

Licence

Online handbook: https://biol2022.github.io/BEDA-handbook

Source: https://github.com/BIOL2022/BEDA-handbook

Updates: https://biol2022.github.io/BEDA-handbook/updates.html

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This handbook has been shaped by the students and staff who have studied and taught Biology Experimental Design and Analysis. Their questions, feedback and practical experience continue to inform each edition.

Contents

1 Unit information

1.1 Welcome

Thank you for joining **BIOL2022: Biology Experimental Design and Analysis**, also known as BEDA. The ability to critically evaluate evidence is a fundamental life skill. In BEDA, we build this skill through hands-on work in experimental design and data analysis. How do we design a robust study? How does a study's design shape its results? How do we choose statistical tools that fit the research question? The reasoning skills you develop will be invaluable not just in biology, but in *any* future career path you choose.

While the focus of BEDA is on the reasoning behind data analysis and interpreting results, we also provide practical guidance on using software. We recommend **Jamovi** if you are new to statistical software. If you already have experience with **R**, you are welcome to use it. **SPSS** and PRIMER are also supported but we prefer to use open-source software where possible as they remain accessible to all.

1.2 Assumed knowledge

BEDA is designed to build on introductory statistics (rather than teach it from the beginning). We use ideas such as data types, sampling variation, hypothesis testing, p-values, confidence intervals and simple statistical models from the opening weeks. This allows us to spend more time on designing biological studies, building appropriate models and interpreting evidence.

Because BEDA has broad prerequisites, qualifying for enrolment into this unit does **not** always mean that you have studied statistics. **However, taking a first-year statistics unit is an assumed knowledge for BIOL2022**. Not fully meeting the assumed knowledge may make BEDA *extremely* challenging as we have limited time to explain basic concepts that are assumed to be known from first-year statistics.

Some enthusiastic students choose to take BEDA straight after their first semester of first year. If this is you, there may be extra challenges, as report writing, group work, and time management skills are usually developed across the full first year of university. For this reason, we strongly recommend taking BEDA in your second year (as intended), so you are better prepared for the workload and expectations of a unit designed for students with more experience.

If you decide to stick around, use Am I ready for BEDA? to see what you need to review so that you can get the most out of the unit.

1.3 Modules

BEDA is organised into three modules, each taught by a different lecturer. Ideas introduced early in the unit continue through the semester as they are applied to different kinds of biological research. Importantly the focus is on the reasoning behind the analysis and how the data are collected, rather than just the analysis itself.

1.3.0.1 Module 1 (Weeks 1–5)

Januar introduces the foundations of study design and statistical modelling. You will look at how a research question, the way a study is designed and the data you collect come together in a model. Some students may be familiar with these ideas from first-year statistics, but we will explore them in a new context. **Importantly, we will use this module to ensure that all students are on the same page with the foundations of study design and statistical modelling, so that we can build on these ideas in later modules**.

The module will introduce you to the following familiar ideas in a new context:

- sampling and how it affects the conclusions you can draw
- study design and variables
- graphical models
- the general linear model (GLM)
- model assumptions of the GLM
- transformations and why we use them
- control variables and random effects as opposed to procedural controls

These ideas help you begin designing your experiment for Report 1.

1.3.0.2 Module 2 (Weeks 6–8)

Clare develops the modelling workflow through univariate experiments, where the analysis focuses on one response variable. You will examine experiments involving categorical predictors, continuous relationships and binary responses, then consider how to choose between possible models.

The module will see your starting to collect data for Report 1. You will also be coached on how to write a report that clearly communicates your study design, analysis and interpretation of results. Additional concepts introduced in this module include:

- binary responses
- non-normal data and dealing with it
- picking the right model and choosing between models

1.3.0.3 Module 3 (Weeks 9–12)

Mat extends the same reasoning to multivariate data, where several response variables are considered together. You will learn how ordination methods such as principal component analysis and multidimensional scaling can reveal patterns in complex datasets, how clustering identifies similar observations, and how multivariate tests evaluate evidence for group differences. **Importantly, you will learn that multivariate methods are great for exploring patterns that you see in the world around you (and are thus very popular in biology)**.

Report 2 brings these ideas together. You will design a multivariate study, collect a group dataset and use it to describe, test and interpret patterns in your environment. We will cover:

- the reason behind multivariate methods
- the ordination methods PCA and MDS
- clustering
- multivariate tests for group differences using PERMANOVA
- ANOSIM and SIMPER for post-hoc analysis

1.4 Lectures & pracs

There are two lectures each week. Unfortunately we do not get to pick the locations, so they are held in different buildings on different days. You are expected to attend 80% of lectures in one way or another. We record this attendance in-person and online when you watch the recordings.

- **Tuesday, 10–11 am** — Belinda Hutchinson Building, Lecture Theatre 1110
- **Wednesday, 10–11 am** — Sydney Nanoscience Hub, Lecture Theatre 4002 (Messel)

You will also attend one two-hour practical each week in your allocated class:

- **Tuesday, 2–4 pm or 4–6 pm**
- **Wednesday, 2–4 pm or 4–6 pm**
- **Friday, 2–4 pm or 4–6 pm**

All practicals are held in **Carslaw Building, Wet Lab 307**. Check your personal timetable for your allocated class and any timetable changes. Do note that because we are a large cohort, once you are allocated to a practical class, you cannot change it. To change your practical class, you must contact the unit coordinator and provide a valid reason, normally a time table clash with another unit or caring responsibilities.

Attendance: You are expected to participate in at least 80% of timetabled activities and must attend at least 10 of the 12 practicals. See **Attendance** under the Code of Conduct for further details.

1.5 Code of conduct

A code of conduct is a shared agreement about how we learn and work together.

BEDA should be a respectful and inclusive place to learn. Treat classmates and staff with courtesy, including when you disagree. Use people's correct names and pronouns, respect different backgrounds and levels of experience, make space for others to contribute, and avoid disrupting lectures or practicals. Discrimination, intimidation and harassment are not acceptable. Follow staff instructions and all laboratory safety requirements when working in laboratory spaces.

1.5.0.1 Working with your group and data

Contribute fairly to your group's work and raise participation problems early. As with any group work, the issues that arise are often best resolved by talking to your group members first. It is important to use a group charter to clarify expectations and responsibilities including how to share data and work together.

Share group data and the information needed to interpret it promptly. Never fabricate data, or omit or alter observations without documenting and justifying the decision. You may discuss your analysis with your group, but any individual report must be written and submitted independently.

More information on how to work work with your group is provided with the Report 1 and Report 2 instructions. If you are unsure about what is acceptable, ask your practical demonstrator or the unit coordinator.

1.5.0.2 Attendance

Attend lectures live where possible. If you miss a lecture, watch the recording promptly. You are expected to have attended or watched the Tuesday lecture before your practical. Because practicals run on Tuesday, Wednesday and Friday, the Wednesday lecture is not assumed knowledge for that week's practical, but you should attend or watch it during the same week.

You must attend at least 10 of the 12 practicals to meet the 80% attendance requirement. Practical attendance is not managed through special consideration. Record your own attendance during each practical using the QR code or website link, then check that it has been recorded correctly. You have one week to request a correction.

1.5.0.3 Raising a concern

For routine group-work, data-sharing or class concerns, speak with your practical demonstrator or the unit coordinator as early as possible. You do not need to approach the teaching team first if the matter is serious or you would be uncomfortable doing so. The University provides separate pathways for complaints, bullying, harassment and discrimination and confidential support through Safer Communities for sexual harm or gender-based violence. In an immediate emergency, call 000 and follow the University's emergency and campus safety guidance.

2

Week 1: Getting started

Biology Experimental Design and Analysis (BEDA)

Introduction

Welcome to your first BEDA practical. Imagine that this is the first day of your new job. A decent workplace environment would see you being introduced to the people, the tools and the resources you will use for your job. Maybe you have done it before, but there is always something new. This practical is *that* introduction.

This week, you will get familiar with the tools and approaches you will use throughout the semester. You will also have time to explore the software and support resources before completing your first modelling task, which is to communicate your ideas using a model, even when you do *not* yet have a statistical model to fit.

Before the practical

1. Attend or watch the Week 1, Tuesday lecture.
2. Read through this practical before class. You will have a better experience if you are familiar with the tasks. **You do not need to complete anything in advance, but reading ahead will help you plan your time and know what to expect.**
3. Bring a laptop. If possible, install your chosen software and the required modules or packages before the practical. If setup is not finished when class begins, work with a partner while the downloads continue.
4. Bring a pen or pencil for your lab notebook. You will use it to sketch ideas, annotate figures, and quickly record decisions during activities.

ⓘ **Note:** Please avoid asking AI to summarise this practical before you have read it yourself. You need to develop critical reasoning skills, which often requires engaging directly with the material and tasks. It takes about 5 minutes to read, and doing so will help you understand what you do (and do not) know so that you can ask for the right help during the practical.

i Download the Week 1 datasets

Download both files before you begin and keep their filenames unchanged:

- Download **week01-penguins.csv**
- Download **week01-possums.xlsx**

If the links do not work, you can also download the files from Canvas - Module Content and Resources.

Learning outcomes

By the end of this practical, you should be able to:

Learning outcome	How you will practise it
Formulate a **measurable** biological question from observations or available data.	Turn observations about penguins into questions, then formulate your own question using variables from another dataset on possums.
Distinguish response and predictor **variables** and classify them by data type.	Classify variables, form biologically meaningful pairs and identify what you want to explain and what might explain it.
Explain how decisions made during **study design** shape the models that can be used.	Compare age recorded as categories with age measured continuously, then consider how measuring or classifying your own variables differently would change the model.
Use a **graphical model** to communicate an expected biological relationship *before* analysing data.	Choose and justify an appropriate plot, label both axes and sketch a plausible relationship in your lab notebook.

2.1 Workshop

We will begin with a short workshop to help you get started.

2.1.1 Pick your software (15 min)

If you are new to statistical software, please choose Jamovi. Demonstrators with expertise in R and SPSS will also be available, although we recommend Jamovi if you would otherwise choose SPSS.

ⓘ **Note:** If you use other software, we assume that you are already familiar with it and can complete the practical tasks independently.

2.1.1.1 Jamovi

1. Go to the Jamovi download page.
2. Download the current **Desktop** version for your operating system.
3. Open the downloaded installer and follow the prompts. On macOS, open the `.dmg` file and move Jamovi to your Applications folder.
4. Open Jamovi.
5. Click the + icon in the top-right corner and select **jamovi library**.
6. Select **Available**, then install **GAMLj3**, **vijPlots** and **flexplot**.

2.1.1.2 R/RStudio

R and RStudio are separate programs. **Install R first, then install RStudio.**

1. Download R for Windows or macOS. On a Mac, choose the installer that matches Apple silicon or Intel.
2. Open the R installer and accept the default options.
3. Download the free version of RStudio Desktop for your operating system.
4. If you cannot install RStudio, you can use Posit Cloud in a web browser or install Positron instead.

2.1.1.3 R/Positron

R and Positron are separate programs. **Install R first, then install Positron.**

1. Download and install R for Windows or macOS. Positron requires R 4.2 or later.
2. Go to the Positron download page.
3. Choose the installer for your computer. On a Snapdragon-based Windows 11 laptop, choose **Windows arm64**. If you do not have administrator access, choose the Windows **user-level** installer.
4. Install and open Positron. If prompted, select the R installation you just installed.

2.1.2 A simple modelling exercise (10 min)

What is in a question? When we see an interesting pattern and want to investigate it, we need to be able to ask questions that can be addressed with data. Let's start with a familiar example.

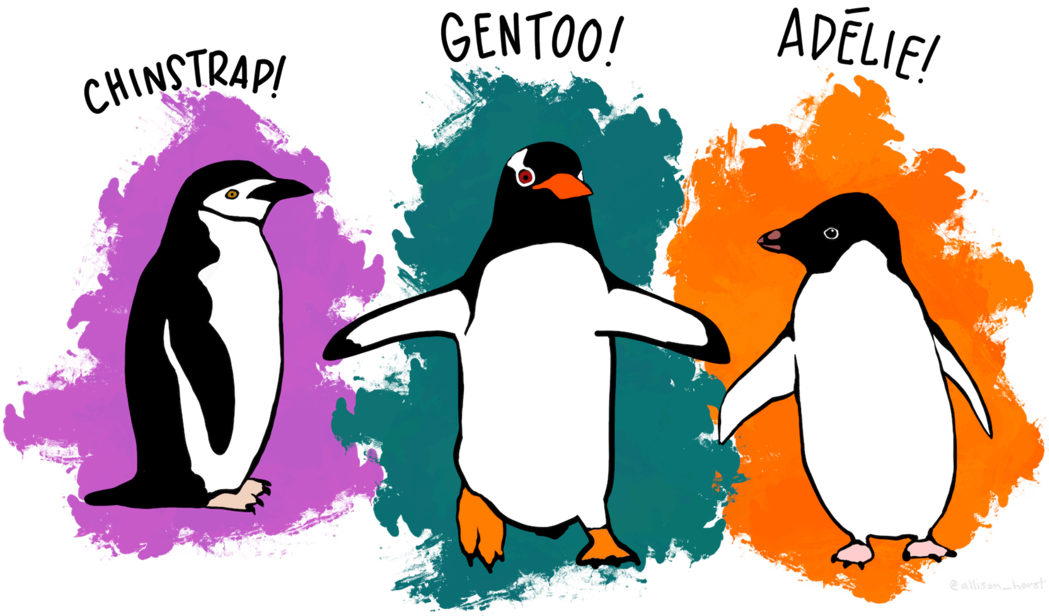


Figure 2.1: The penguin dataset. Artwork by Allison Horst.

The above are simple illustrations of the three penguin species in the `week01-penguins.csv` dataset. Suppose we were, in fact, scientists on a research expedition to the Palmer Archipelago of Antarctica, and we saw these cartoon penguins. *Imagine these are the only observations available to us: the penguins are not moving, so we can only compare their shapes and sizes, and we can see at least three species.* What interesting observations might we make about them? How could we turn those observations into a question that can be addressed with data?

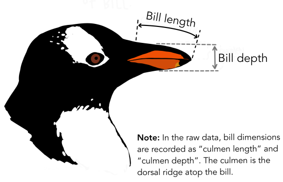


Figure 2.2: Bill dimensions in penguins. Artwork by Allison Horst.

To work with measurable data, you need to identify the **response** and **predictor** variables in a question. What is a response variable? What is a predictor variable? Write your own definitions in your lab notebook, as you will use these terms throughout the semester.

Next, we will open the `week01-penguins.csv` dataset and discuss variables, data types, relationships, and how to formulate questions. For this activity, *pretend* you are biologists who have just observed these penguins in the wild and are deciding which variables are important to measure. **We will work together to formulate a question and sketch a model of the relationship between them. No software is needed.**

2.1.3 Introduction to cheatsheets (5 min)

In the final part of the workshop, we will introduce cheatsheets, which are quick reference guides that we have developed to guide you through the software. We hope that this makes the purpose of BEDA clear: software should be the *least* of your worries.

2.2 Practical

You will now work in pairs to complete the practical exercises. Your demonstrators will roam the room and check in with you to see how you are going. If you have any questions, please ask them! **Demonstrators will check each of your exercises to ensure that you are on the right track.**

The times below are a guide. You must complete Exercise 3 in full, so ask for help early if software setup delays you.

2.2.1 Exercise: Cheatsheets (25 min)

This exercise is designed to get you familiar with the cheatsheets and how to use them. You will be asked to complete a few cheatsheets that are based on Jamovi or R. Please do not attempt other cheatsheets for now - we want to make sure you are familiar with the software today and catch technical issues early.

2.2.1.1 Task 1

1. Open Cheatsheets.
2. Choose **two cheatsheets** that you have not used before and see if you can create the outputs they describe.
3. Save those outputs. A demonstrator will check your work and give you feedback.

ⓘ **Note:** If you have never used Jamovi before, now is a good time to try it.

2.2.2 Exercise: Data types (25 min)

In this exercise you will work on the **week01-possums.xlsx** dataset. If you have not already downloaded it, do that now and open it in your chosen software.

2.2.2.1 Task 2

1. Carefully read the “Metadata” and “AllPossData” sheets in the MS Excel file. The metadata sheet describes the variables in the dataset and their units of measurement. The AllPossData sheet contains the actual data.
2. Choose six variables, including at least two continuous variables and two categorical variables. Classify each variable and record its type in your lab notebook. Some variables may have more than one reasonable classification.

3. Use those variables to identify two biologically meaningful pairs: one containing a continuous and a categorical variable, and another containing two continuous variables. You will choose one of these pairs for Task 3.

ⓘ **Note:** The same underlying trait can be recorded in different ways. For example, exact age can be treated as continuous, whereas age classes such as young, adult and old are ordinal. Binary variables are also categorical variables with exactly two levels. If you are unsure about a classification, ask a demonstrator. Data classification is a key step in the modelling process.

Data types

Data type	What it means
Continuous	A numeric measurement that can take many values
Discrete	A numeric count with whole-number values
Categorical (nominal)	A label or group with *no natural order*
Ordinal	A category with a *meaningful* order
Binary	A categorical variable with *two levels*

2.2.3 Exercise: Modelling basics (45 min)

In this exercise you will use variables from the **week01-possums.xlsx** dataset to formulate one biological question and develop one graphical model. You will choose an appropriate plot and sketch the relationship you expect to see; you do not need to analyse the data in software.

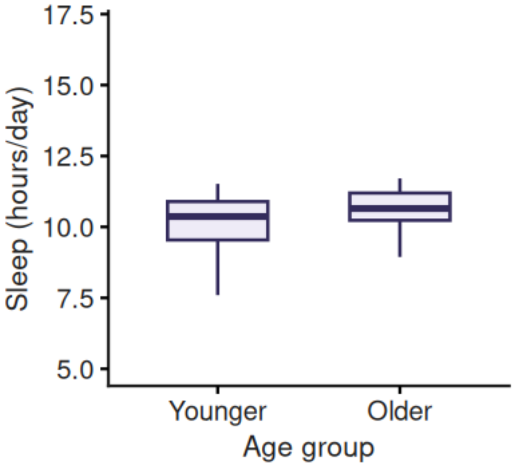
What is the point? A model is just a simple way to represent something complicated. In data analysis, a model helps us make sense of a dataset, test ideas (hypotheses), or make predictions. Models can be as simple or as fancy as you like, depending on your data and your research question.

Soon, you will use empirical models to describe relationships observed in data. We write the structure of these models using simple notation, such as $y \sim x$, where y is the response variable and x is the predictor variable. We won't fit a statistical model just yet; for now, we will use plots to represent relationships we might expect to see.

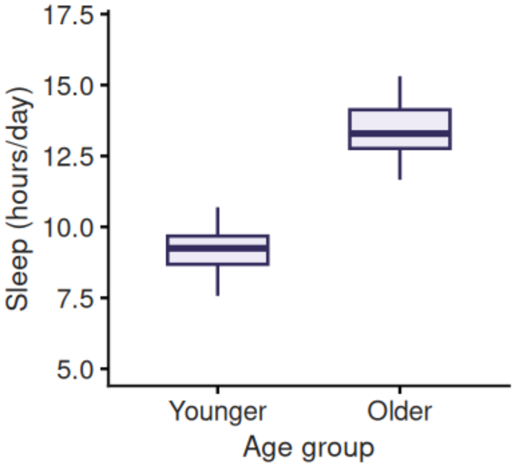
2.2.3.1 If you can plot the relationship, you can begin to model it

For now, we will use plots as simple graphical models. They are abstractions: simplified representations of relationships we expect to see between variables. These are not fitted statistical models yet; they are a way to make our expectations explicit.

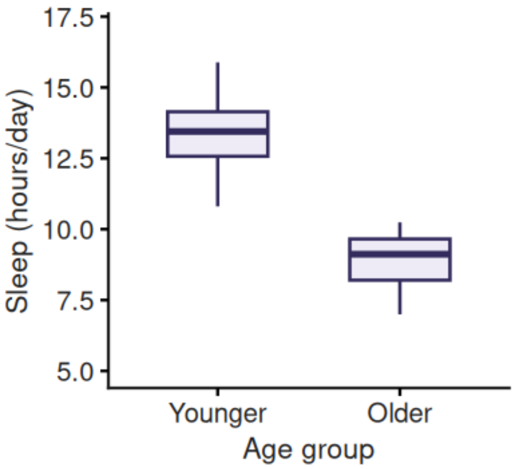
For example, you could plot the daily sleep time of dogs against their age to see if there is a pattern. **At this stage, more than one pattern may be biologically plausible. The important thing is to make your expectation explicit and be able to explain it.** Younger and older dogs might sleep for similar amounts of time, or one group might sleep longer than the other. These possibilities could look like this:



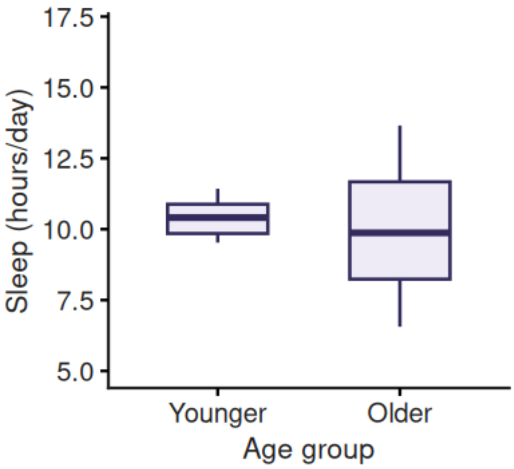
(a) Similar sleep time between age groups.



(b) Older dogs sleep longer.



(c) Younger dogs sleep longer.



(d) Greater variation in sleep time among older dogs.

Figure 2.3: Four possible patterns in daily sleep time between younger and older dogs.

Each plot represents a different possible relationship, but they all use the same response and predictor variables. The response is daily sleep time in hours, and the predictor is age group (younger or older)—a categorical variable.

Interestingly, how we consider our variables can drastically change the type of plot and model used for data analysis. For example, consider the same relationship as above, **but while planning the study, we decide to record the exact age of each dog in years**. Now, age is a continuous variable, and we can use a scatterplot to show how daily sleep time changes across the age range:

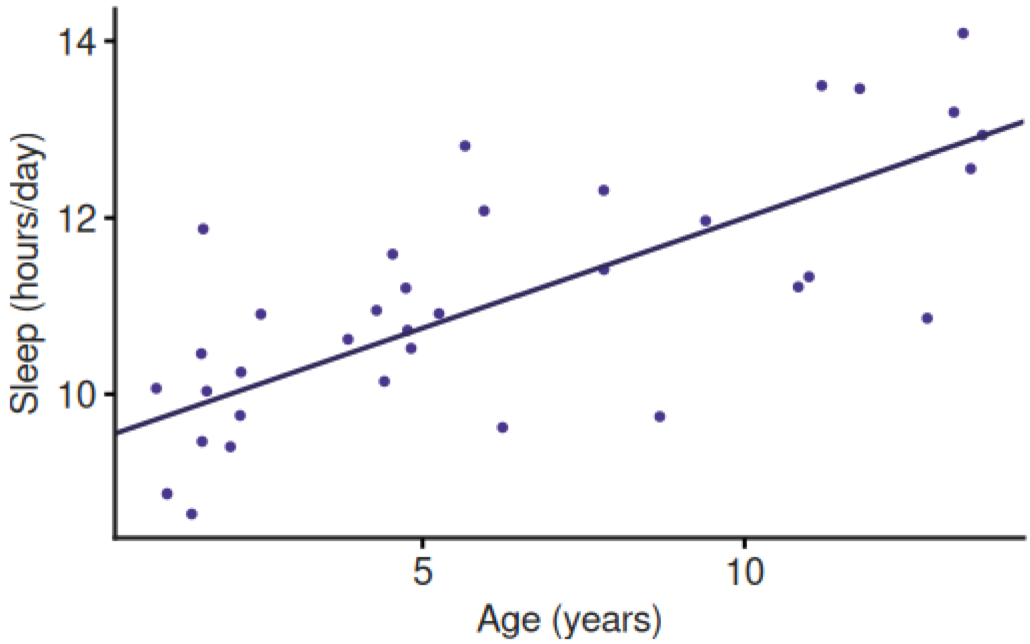


Figure 2.4: One possible relationship between daily sleep time and age in dogs.

In the scatterplot, the exact ages are retained, making it easier to see how daily sleep time changes across the age range. The boxplots and scatterplot use the same response and predictor, but treating age as categorical or continuous changes what the plot shows and how we interpret the relationship.

2.2.3.2 Task 3

Your goal is to practise thinking like a research biologist. Complete all five steps below for **one biological question and one graphical model**. You do not need to process or analyse the data. Knowing the variable types is enough to choose and sketch an appropriate plot.

1. **Choose a variable pair.** Return to the two pairs you identified in Task 2 and choose one for this task.
2. **Formulate a research question.** Choose variables that could address a simple biological question. Identify the response variable, the predictor variable and the type of each variable.
3. **Choose a plot.** Match the plot to your question and variable types:
    - **Scatterplot:** Shows the relationship between two continuous variables.
    - **Boxplot:** Compares a continuous response across the groups of a categorical predictor.
4. **Sketch your graphical model.** In your lab notebook, write your research question and draw the plot you expect to see. Label both axes and show a plausible pattern. The sketch represents your idea; it does not need to match the data.
5. **Explain your model.** State the relationship in words and explain why the plot suits your variables. Consider whether a different way of measuring or classifying one variable would change the plot you use. Discuss your reasoning with your partner or demonstrator.

Your lab notebook should contain:

- your biological question;
- the response and predictor variables, including their types;

- a labelled sketch of the plot;
- the expected relationship stated in words; and
- a brief explanation of why the plot is suitable.

You do not need to fit or test a statistical model.

2.3 End of practical notes

2.3.1 Help us collect lots of snails!

We need about 400 common garden snails for the Module 2 plant-herbivore projects! Lots of prizes in gift cards to be won. See the snail competition page for details. We have prepared **up to 17 prizes** for this competition, so check out the Canvas page and help us.

If you are interested, please collect the following from the lab before you leave:

1. A food container
2. A pair of gloves (latex or nitrile)

Taking part is optional and has no effect on your marks.

2.3.2 That is a wrap

That's it for today! If you have any questions, ask your demonstrators who are happy to help. If there is time, we will do a 10-minute Q&A session before showing the attendance QR code. Make sure to take attendance before you leave.